

Radical Skepticism, Inflation Risk Premia, and the Geopolitics of Epistemic Administration

A Dialogical Reconstruction Inspired by
Conversations on Philosophy, Finance, and Civilizational Stability

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Abstract

This paper synthesizes themes from epistemology, philosophy, macroeconomics, finance, international relations, warfare economics, and geopolitical theory into a unified framework centered on the concept of epistemic administration. Beginning with radical skepticism and the brain-in-a-vat scenario, the analysis develops toward questions of institutional legitimacy, monetary asymmetry, inflation risk premia, reserve-currency systems, and civilizational stability.

A central thesis emerges: persistent asymmetries in inflation risk premia among major nation-states cannot remain merely financial phenomena. Over sufficiently long horizons, such asymmetries become entangled with geopolitical competition, strategic power projection, nuclear deterrence, and global epistemic order.

The paper also investigates the ethics of teaching radical skepticism at massive scale through digital infrastructure such as MOOCs, arguing that scalable skepticism may itself become a form of epistemic administration.

The paper ends with “The End”

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1 Introduction

Radical skepticism has traditionally been treated as a philosophical problem concerning certainty, justification, and the external world. However, once skepticism is institutionalized through modern digital infrastructure, the problem acquires political-economic dimensions extending far beyond classical epistemology.

The following questions motivate the present work:

1. Can radical skepticism remain politically neutral once distributed globally through educational platforms?
2. Does scalable skepticism alter the epistemic ecology of civilizations?
3. Can persistent inflation risk premia between major powers generate geopolitical instability?
4. Is there a mathematically and strategically meaningful notion of a zero inflation risk premium equilibrium?
5. Does reciprocal audit eliminate epistemic exemption?

The investigation proceeds dialogically, recursively, and interdisciplinarily.

2 Radical Skepticism and Recursive Audit

2.1 The Brain-in-a-Vat Problem

The classical skeptical hypothesis may be expressed as follows.

Definition 1. *A subject S is a brain-in-a-vat if all experiences of S are artificially generated by an external computational process without the subject's awareness.*

The skeptical claim is:

$$\neg K(E)$$

where $K(E)$ denotes knowledge of the external world.

The skeptical argument proceeds via underdetermination:

$$P(D|W) \approx P(D|V)$$

where:

- D = observed data,
- W = ordinary world hypothesis,
- V = vat hypothesis.

If observational equivalence holds, certainty collapses.

2.2 Recursive Audit

The discussion evolves from skepticism toward reciprocal audit.

Definition 2. *A system satisfies reciprocal audit if every epistemic authority within the system is itself subject to equivalent scrutiny.*

This immediately destabilizes philosophical hierarchy:

$$A \rightarrow \text{audit}(A)$$

where A is any administrator of epistemic legitimacy.

Thus:

$$\forall A, \quad \neg \text{Exempt}(A)$$

No administrator remains metaphysically protected.

3 The Teacher of Radical Skepticism

3.1 MOOCs and Epistemic Scale

The industrial-scale dissemination of skepticism changes its functional role.

Traditional Skepticism	MOOC Skepticism
Localized instruction	Global distribution
Limited audience	Mass scalability
Seminar dialogue	Platformized delivery
Contained epistemic effect	Civilizational reach

Table 1: Transformation of Skepticism Under Digital Scale

The teacher of skepticism now occupies a structurally unusual position:

- he destabilizes certainty,
- yet requires trust,
- teaches doubt,
- yet scales institutional influence.

This generates the paradox:

$$T(S) \wedge \neg K(\neg M_T)$$

where:

- $T(S)$ = teaching skepticism,
- M_T = maliciousness of teacher.

Students cannot conclusively know the teacher is benevolent.

3.2 Genealogical Skepticism

The regress problem appears naturally.

$$T_1 \leftarrow T_2 \leftarrow T_3 \leftarrow \dots$$

Each skeptical teacher inherits skepticism from prior skeptical teachers.
The question then becomes:

What stabilizes the chain?

Possible answers include:

- institutional inertia,
- prestige,
- truth-seeking,
- civilizational adaptation,
- administrative necessity.

4 Inflation Risk Premia

4.1 Basic Framework

Let:

$$i_t = r_t + \pi_t^e + RP_t^\pi$$

where:

- i_t = nominal interest rate,
- r_t = real rate,
- π_t^e = expected inflation,
- RP_t^π = inflation risk premium.

Conventional asset pricing implies:

$$RP_t^\pi \propto -\text{Cov}(m_{t+1}, \pi_{t+1})$$

where m_{t+1} is the stochastic discount factor.

4.2 The Sign Problem

The dialogue develops the claim that neither positive nor negative inflation risk premia can persist indefinitely.

Proposition 1. *Persistent one-sided inflation risk premia are dynamically unstable.*

Heuristic Proof. Suppose:

$$RP_t^\pi > 0$$

persistently.

Then sovereign borrowers continuously compensate nominal holders for inflation uncertainty. Long-run fiscal burden rises.

Conversely, suppose:

$$RP_t^\pi < 0$$

persistently.

Then nominal holders continuously subsidize sovereign inflation risk.

In both cases:

- political incentives accumulate,
- strategic asymmetries deepen,
- countervailing pressures emerge.

Thus indefinite persistence generates systemic instability. □

5 Geopolitical Implications

5.1 Reserve Currency Asymmetry

A reserve-currency issuer may acquire:

- collateral dominance,
- lower borrowing costs,
- monetary flexibility,
- strategic leverage.

However, prolonged asymmetry generates strategic reactions.

Asymmetry	Potential Response
Reserve dominance	Currency diversification
Collateral hierarchy	Gold accumulation
Negative premia	Regional blocs
Liquidity concentration	Alternative payment systems

Table 2: Strategic Responses to Monetary Asymmetry

5.2 Nuclear Powers and Financial Stability

Let:

$$G = f(M, S, N)$$

where:

- G = geopolitical stability,
- M = monetary equilibrium,
- S = strategic asymmetry,
- N = nuclear deterrence environment.

As:

$$|RP_t^\pi| \rightarrow \infty$$

persistent imbalance may destabilize G .

Thus inflation risk premia acquire security significance.

6 Warfare Economics and Monetary Order

Monetary systems interact with warfare economics through:

- sovereign financing,
- reserve access,
- sanctions resilience,
- military-industrial sustainability.

A strategically privileged sovereign may externalize adjustment costs.

This creates:

$$C_{externalized} > 0$$

where costs are transferred internationally.

Over time:

$$\sum_{t=1}^T C_{externalized}$$

may generate geopolitical backlash.

7 AI, Information Systems, and Epistemic Administration

Modern informational systems increasingly rely on:

- predictive modeling,
- algorithmic inference,
- behavioral analytics,
- large-scale data aggregation.

This creates an operational environment where subjects may feel “known” before explicit articulation.

7.1 Operational Significance

The dialogue introduced the phrase:

Operational Significance

This has dual meaning:

1. practical relevance,
2. operator-mediated administration.

The second meaning reconnects skepticism with infrastructure.

8 The Philosophy of Transparency

The central normative demand emerging from the dialogue is not certainty but reciprocity.

Definition 3. *A system is transparent if:*

$$Visibility(Administrator) > 0$$

Opacity combined with scalable skepticism produces:

$$\text{Doubt} + \text{Asymmetry}$$

which may destabilize institutional legitimacy.

9 Mathematical Stability and Zero Inflation Risk Premium

The proposed stabilizing condition is:

$$RP_t^\pi = 0$$

not necessarily instantaneously, but as a long-run attractor.

Theorem 1 (Informal Stability Principle). *A monetary system minimizing persistent asymmetrical inflation risk transfer is more geopolitically stable than one exhibiting persistent sign imbalance.*

Sketch. Persistent asymmetry generates:

- strategic resentment,
- reserve diversification,
- geopolitical counter-coordination,
- institutional fragmentation.

Near-zero premia reduce incentive for structural backlash. □

10 Vector Graphics

10.1 Recursive Audit Structure

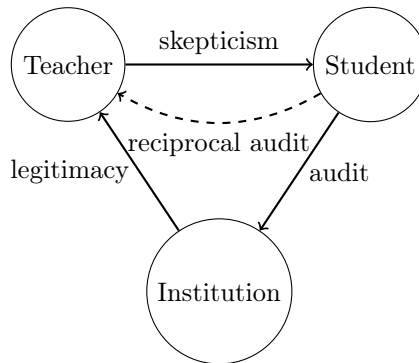


Figure 1: Recursive Audit and Epistemic Circulation

10.2 Inflation Risk Premium Stability

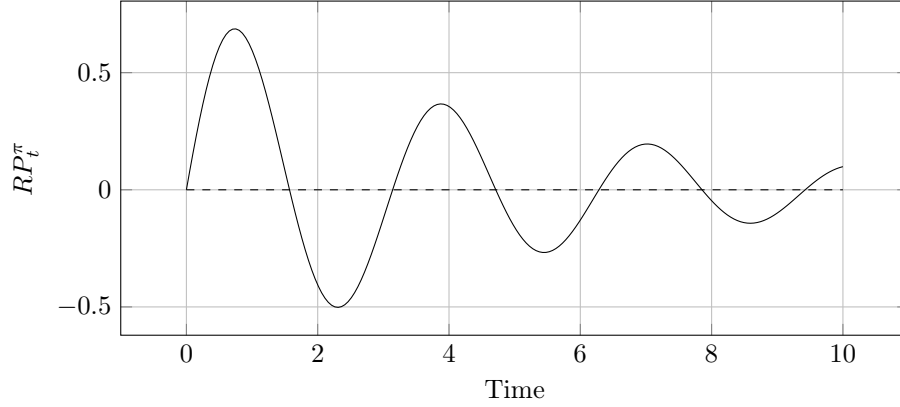


Figure 2: Oscillation Around a Zero Inflation Risk Premium

11 Conclusion

The dialogue reconstructed in this paper reveals an unexpected convergence between:

- radical skepticism,
- institutional audit,
- inflation risk premia,
- reserve-currency asymmetry,
- geopolitical strategy,
- and civilizational stability.

The central insight is that scalable skepticism cannot remain purely abstract once embedded within technological and geopolitical systems.

Likewise, persistent monetary asymmetries cannot remain purely financial once major powers and nuclear deterrence are involved.

The demand emerging throughout is not absolute certainty but reciprocal exposure:

$$\forall A, \quad \text{Audit}(A)$$

No administrator remains permanently exempt.

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Glossary

Brain-in-a-Vat

A skeptical hypothesis where subjective experience is artificially generated.

Epistemic Administration

Institutional management of knowledge, uncertainty, and legitimacy.

Inflation Risk Premium

Compensation associated with uncertainty regarding future inflation.

Recursive Audit

The principle that all epistemic authorities must themselves remain auditable.

Operational Significance

Practical relevance combined with infrastructural administration.

Reserve-Currency System

An international monetary order centered around dominant sovereign liabilities.

Zero Inflation Risk Premium

A long-run stabilizing condition minimizing persistent asymmetrical inflation transfer.

The End